

LESSON 3.8

SOLVING REAL-WORLD PROBLEMS WITH RATIONAL NUMBERS – PART 4



LESSON INTRODUCTION

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

LEARNING TARGETS

- I can solve problems with at least three operations involving any of the four operations with rational numbers.

BENCHMARK COHERENCE

BUILDING ON

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

WORKING TOWARD

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

ESSENTIAL QUESTIONS

- How do you solve multi-step real-world problems involving any of the four operations with rational numbers?

LESSON NARRATIVE

This lesson applies skills students have used throughout the unit to solve various real-world scenarios using operations with rational numbers. Students work in small groups to analyze and answer questions about a given scenario, then present their work to the rest of the class through a gallery walk. This lesson culminates in a self-reflection on student understanding of using operations with rational numbers.

COMPONENT SUMMARY

3.8.1 WARM-UP (~5 MIN.)

Analysis of image of office supplies

3.8.2 ACTIVITY (25 MIN.)

Gallery Walk – exploring real-world contexts for problem-solving

3.8.3 YOUR TURN! (10 MIN.)

Practice performing operations with rational numbers

3.8.4 WRAP-UP (~5 MIN.)

Self-reflection on level of understanding

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Balance

MATERIALS

- Chart paper
- Calculator
- (Optional) Ruler

ADDITIONAL PREPARATION

- Gallery Walk – provide each scenario on chart paper for students to rotate while exploring. For smaller group sizes, make 2 copies of each scenario, splitting the class into “A” and “B” rotations in which each letter contains all four scenarios.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may use the incorrect operation(s) to reach the answer
- Students may misunderstand vocabulary used in the real-life scenarios.

THINGS TO CONSIDER

- Consider how students can add creative elements to the gallery walk. Encourage students to utilize diagrams, tables, pictures, or other creative elements. Students can use construction paper to showcase different elements.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

The 3.8.1 Warm-Up presents a “Notice and Wonder” routine. Students will examine the image and will state what they notice and what they wonder about from the image presented.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

This standard emphasizes experiencing math through real-world examples. Students will be exploring various real-world examples that use various forms of rational numbers and will perform all operations. Provide opportunities for students to make connections between each context and their personal experiences.

DIFFERENTIATE INSTRUCTION

Consider providing translations of each context for English Language Learners, or adjust with cultural intelligence in mind, affording students in the early stages of language acquisition development the opportunity to create their own contexts and problems based on their experiences and cultural capital.

Additionally, offer all students a variety of reading comprehension strategies to employ during the Gallery Walk, making use of tools and resources for analyzing and unpacking each context.

- Three Reads: Instructional routine for targeting specific information
- Mark-Up Text: Visual strategy for identifying important information
- Partner Read: Auditory scaffolding for students to engage verbally in supporting different entry points

3.8.1 WARM-UP: NOTICE AND WONDER

TEACHER MOVES

Notice and Wonder

The Warm-Up presents an opportunity for students to use mathematical language in order to communicate suitability of different rational numbers in real-life situations. Engage the class in a Notice and Wonder routine to foster discussion. Students may notice that the width of each paper clip could be described using decimals. Students might comment that fractions of a ream of paper could be used to communicate how many sheets of paper each binder clip can hold. Students might also state that negative rational numbers are not applicable in this situation.



3.8.1 Warm-Up: Notice and Wonder

1. What do you notice about the binder clips in the image?

Answers will vary. There are three sizes of binder clips pictured – small, medium, and large. The binder clips are in order from right to left and the largest clip is on the right and they are going down in size to the left.



2. What questions come to mind when you look at this image?

Answers will vary. Which binder clip can hold the stack of paper pictured? What is the maximum number of pages that each binder clip can hold?

3. The stack of papers contains about 260 sheets of paper. Estimate the number of sheets each binder clip can hold.

Answers will vary.

<i>Small Binder Clip</i>	<i>Medium Binder Clip</i>	<i>Large Binder Clip</i>
<i>75</i>	<i>175</i>	<i>275</i>

4. To make a good estimate, what information would be useful to know?

Answers will vary. How many inches does each binder clip hold? How many inches is a sheet a paper?

5. What tools or resources could help you find that information?

Answers will vary. One tool that would be helpful with finding the information is a ruler to measure the width of the binder clip opened.



3.8.2 Activity: Gallery Walk

Your teacher will assign your group one of the following scenarios. Work together to answer the question(s), writing neatly on the chart paper provided. Show all your work to prepare a presentation for the other students in your class.

Scenario 1

Marie plans to use a ride sharing service for the trip from Tampa International Airport to her hotel. The service has a booking fee of \$2.45 and a cost of \$0.42 for the first $\frac{1}{4}$ mile and an additional fee of \$0.16 for each 0.1 mile after the first $\frac{1}{4}$ mile.

1. How much will the trip cost if Marie's hotel is 7.25 miles from the airport?

$$\$2.45 + \$0.42 + \left[(\$0.16) \left(\frac{7}{1} \right) \right] = \$14.07$$

2. Marie decides to stop by a grocery store before arriving at her hotel. The ride sharing service charges \$0.45 for each minute they have to wait on a customer. Her total bill was \$21.72. How long was Marie in the grocery store?

$$21.72 - 14.07 = 7.65$$

$$7.65 \div 0.45 = 17 \text{ minutes}$$

3.8.2 ACTIVITY: GALLERY WALK

TEACHER MOVES

MA.K12.MTR.7.1: Real-World Contexts

Model connections to personal experiences for students to actively engage with applications of mathematics. Consider having students answer these questions on their chart paper:

- Is this real-world context something you have experienced?
- What experiences have you had that are similar to this real-world context?

DIFFERENTIATE INSTRUCTION

Gallery Walk

Have small groups of students work one of the assigned questions on chart paper. When all groups are done, post the chart paper around the room. Have students "Gallery Walk" to each question, adding comments or suggested error corrections to the work. If the class is large, make 2 copies of each question for smaller working groups.

ENRICHMENT

What form of rational numbers will each answer be?

3.8.2 ACTIVITY: GALLERY WALK (CONTINUED)

TEACHER MOVES

Students may struggle with whether to include 4 or 5 days in question 1. Ask students which they consider is best and why. Modify your key based on student choice. Inform these students that they should be consistent with their choice as they answer question 2.

Scenario 2

Leah and her parents have an agreement. She borrowed money from them to buy a video gaming system and is supposed to pay them back within 3 months. If Leah does not have the balance paid by the agreed-upon time, she will have to give the system to her parents until it is paid off. Leah and her parents created a payment plan to help her meet this goal. Starting on July 1, Leah should pay \$5.95 per day until the gaming system is paid off. On July 5th, her balance was -\$10.25 since she couldn't pay the full amount each day. Leah was not able to pay anything again for 15 days. On July 20th, Leah paid \$85.00.

1. How much had Leah paid her parents by July 5th?

$$(\$5.95)(5) + (-\$10.25) = \$19.50$$

2. By July 20th, how much more or less had Leah paid than what was required by the payment plan?

$$(\$5.95)(20) + (-\$19.50) + (-\$85) = \$14.50$$

Leah had to pay \$14.50 more on her payment plan.

DIFFERENTIATE INSTRUCTION

Consider capturing the cultural intelligence or cultural capital of your students that are unique to their class and their peers and applying that capital for teaching and learning.

Is there a local coffee shop or tienda to use as an anchor or reference? Consider providing a picture or actual worksheet of their changes to better engage students.

QUESTIONING

Extend student thinking to interpret the changes by day.
Based on the data in this table:

- Which day might the shop be most busy?
- Is there a day the coffee shop may be closed? How do you know?
- Should the shop order more or less on certain days of the week?

A coffee shop keeps track of changes in their coffee inventory. A week of their record-keeping is shown.

Day	Change in Inventory (lb)
Monday	150
Tuesday	-43.3
Wednesday	-50.5
Thursday	72.8
Friday	-52.7
Saturday	-21.6
Sunday	0

Scenario 3

Coffee is delivered in 50-pound (lb) bags, 3 bags at a time.

1. How much coffee is left at the end of the week?

$$54.7 \text{ lbs}$$

2. Coffee is delivered in 50-pound (lb) bags, 3 bags at a time. How many pounds of coffee were sold on Thursday?

$$150 - 72.8 = 77.2 \text{ pounds}$$

Scenario
4

Recall from the previous lesson that The Better Utility Company charges \$0.12 per kilowatt-hour for the energy a customer uses that is provided by the utility company. The company gives an adjustment of $-\$0.025$ for every kilowatt-hour of electricity a customer with solar panel(s) generates that they do not use themselves.

The table shows the number of kilowatt-hours of the electricity a customer used and generated each week for a month.

	Kilowatt-Hours Used	Kilowatt-Hours Generated that were not Used by the Customer
Week 1	112.08	253.2
Week 2	181.5	357.6
Week 3	101	308
Week 4	200.42	214.4

1. What is the amount due for this month?

$$(\$95)(\$0.12) + (1133.2)(-\$0.025) = \$43.07$$

	Used	Not Used	Total
Week 1	112.08	253.2	\$7.12
Week 2	181.5	357.6	\$12.84
Week 3	101	308	\$4.42
Week 4	200.42	214.4	\$18.69

2. The average customer's bill is $\frac{2}{3}$ of what they would have been charged if they had not produced electricity using solar panels. Explain if this customer's savings was below or above that average.

Answers will vary. This customer paid about 60.3% of what they would have paid, which is less than $\frac{2}{3}$ of the original amount, so they saved more than the average customer who has solar panels.

3.8.2 ACTIVITY: GALLERY WALK (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider challenging students to consider sociocultural contexts that could facilitate meaningful discussions.

- Should energy be a free resource provided by the government? Why or why not?
- Should solar energy be required by all homes? Why or why not? What challenges would there be?
- If solar energy costs more money but conserves the environment, is it worth it? Why or why not?

3.8.3 YOUR TURN!

QUESTIONING

Offer students the option to perform operations using one form of rational numbers. Allow students to convert all mixed numbers into decimals and complete, and also allow students to convert all decimals into fractions or mixed numbers and complete.

DIFFERENTIATE INSTRUCTION

In order to assist students with organizing calculations throughout problem solving for question 2, encourage students to add an additional column on the table given. This column would be an organized visual to record the total price of each item purchased.



3.8.3 Your Turn!

- Araya is making red ribbons to sell to support a local nonprofit organization. She purchased a roll of fabric that is $30\frac{3}{4}$ feet in length and each ribbon uses 0.635 feet of fabric.

If she already has 16 ribbon orders, how many additional ribbons will she be able to sell with the single roll? Show your work.

$$\left(30\frac{3}{4} \div 0.635\right) - 16 = 32.425 \quad 32 \text{ Ribbons}$$

- Joanne purchased $2\frac{1}{4}$ pounds of tomatoes, $3\frac{1}{8}$ pounds of red onions, $\frac{3}{4}$ pounds of asparagus, and 3 lemons from a grocery store.

Produce	Price
Tomatoes	\$2.99 per pound
Red Onions	\$1.79 per pound
Asparagus	\$2.49 per pound
Lemons	\$0.79 each

Based on the listed prices, how much would Joanne spend?

Approximately \$16.56

**3.8.4 Wrap-Up: Reflection**

1. Color in the arrow up to the statement which best describes your current understanding of solving real-world problems with rational numbers.

Answers will vary.



I'm so confident, I could explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and I listened, but I am finding this challenging. I will make sure that I get help.

I do not understand any of this yet. There are things I could do to be a better learner.

3.8.4 WRAP-UP: REFLECTION**DIFFERENTIATE INSTRUCTION**

Consider having students provide examples of their level of understanding using a problem from the lesson. Students who are confident can write an explanation of how they solved. If they're struggling, students can circle or name which part of the problem they have questions about.

Note for growth mindset – avoid students naming things “they are not good at” or “they can’t do.” Language is important; therefore, students instead should use language like “Things I have questions about” or “I need help with.”